

Performance Evaluation of AOMDV Protocol in Wired and Wireless Networks using ns-3

1 Introduction

This report presents a performance evaluation of the AOMDV (Ad hoc On-demand Multipath Distance Vector) routing protocol using the ns-3 network simulator. The study analyzes both wired and wireless network scenarios under varying parameters to understand the behavior of the protocol under different network conditions.

The key performance metrics evaluated include throughput, end-to-end delay, packet delivery ratio (PDR), packet drop ratio, and energy consumption.

2 Network Topologies Under Simulation

Two different network environments were considered:

2.1 Wireless Network

- Multi-hop ad hoc network
- Nodes deployed with mobility
- AOMDV routing protocol used

2.2 Wired Network

- Static topology
- Point-to-point links
- AOMDV applied for routing

3 Parameters Under Variation

The following parameters were varied:

- Number of nodes
- Number of flows
- Packets per second (PPS)
- Node speed (wireless only)

3.1 Note on Parameter Selection

The originally specified parameters were:

- Nodes: 20, 40, 60, 80, 100
- Flows: 10, 20, 30, 40, 50
- PPS: 100, 200, 300, 400, 500

4 Modifications Made in the Simulator

The following modifications were implemented:

- A new **AOMDV routing module** was implemented from scratch under `src/aomdv`, including helper, routing, packet, and routing-table components integrated with the ns-3 IPv4 routing framework.
- Custom control packet headers were implemented for **RREQ**, **RREP**, **RERR**, and **HELLO** messages, with serialization/deserialization support and explicit message typing.
- Multipath routing state was added by storing multiple next hops per destination (with sequence numbers, hop counts, and expiry times), selecting the best valid path for forwarding, and purging expired paths periodically.
- Route discovery/maintenance logic was implemented with duplicate RREQ suppression, reverse-path installation from RREQs, forward-path installation from RREPs, and route error propagation/invalidations when links fail.
- Per-interface UDP sockets and timers were used to exchange AOMDV control traffic, with configurable attributes such as maximum paths, active route timeout, hello interval, and path discovery time.

5 Results

5.1 Wireless Network Results

5.1.1 Effect of PPS

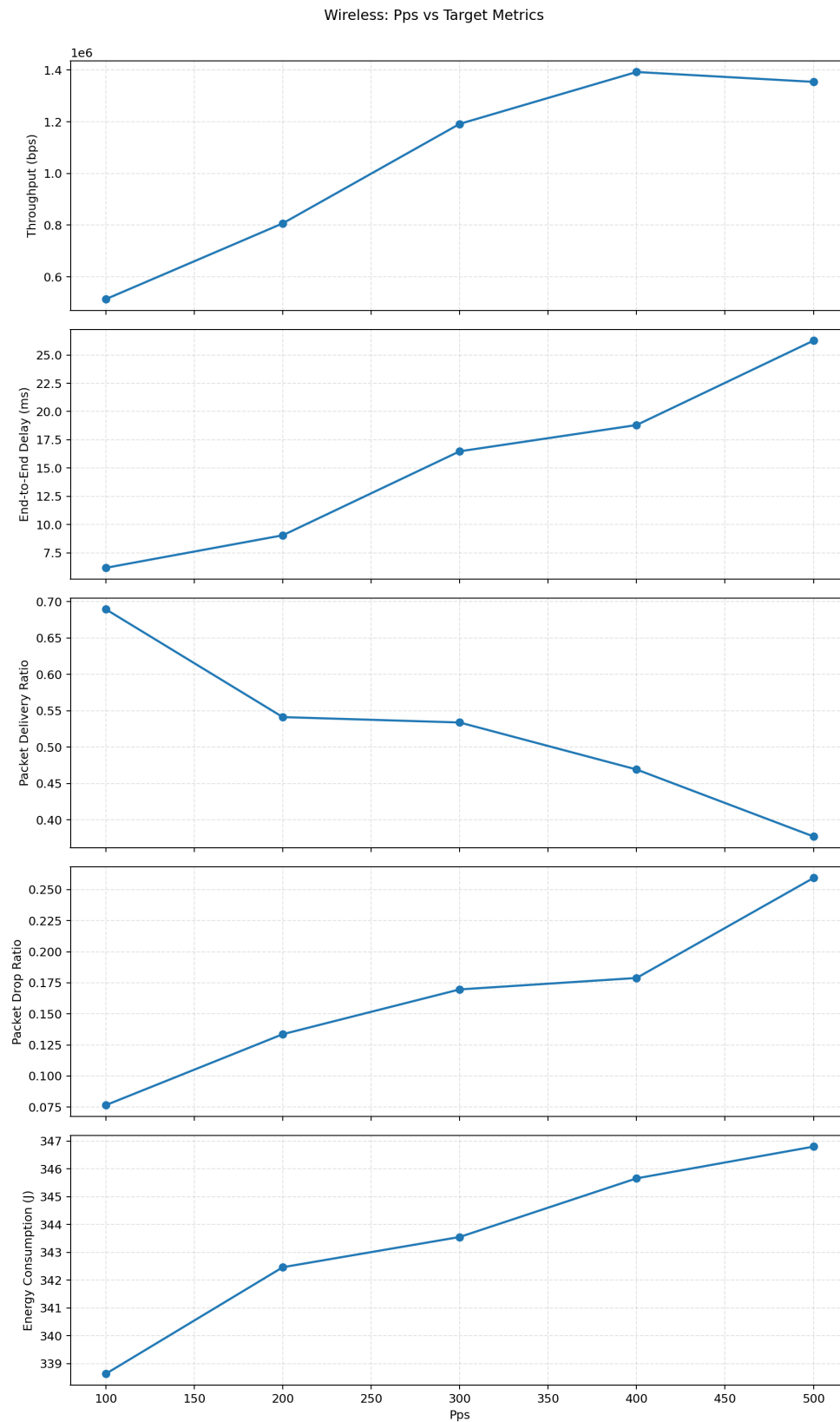


Figure 1: Wireless: Performance vs PPS

Throughput increases with PPS initially, reaching a peak before slightly declining due to network congestion. Delay and packet drop ratio increase steadily with higher PPS, while PDR decreases. Energy consumption also increases with load.

Table 1: Wireless PPS sweep results

PPS	Throughput (bps)	Delay (ms)	PDR	Drop Ratio	Energy (J)
100	513432	6.18	0.6896	0.0766	338.63
200	806760	9.06	0.5414	0.1336	342.46
300	1191670	16.48	0.5340	0.1697	343.55
400	1392550	18.81	0.4694	0.1789	345.66
500	1354320	26.27	0.3773	0.2593	346.80

5.1.2 Effect of Number of Flows

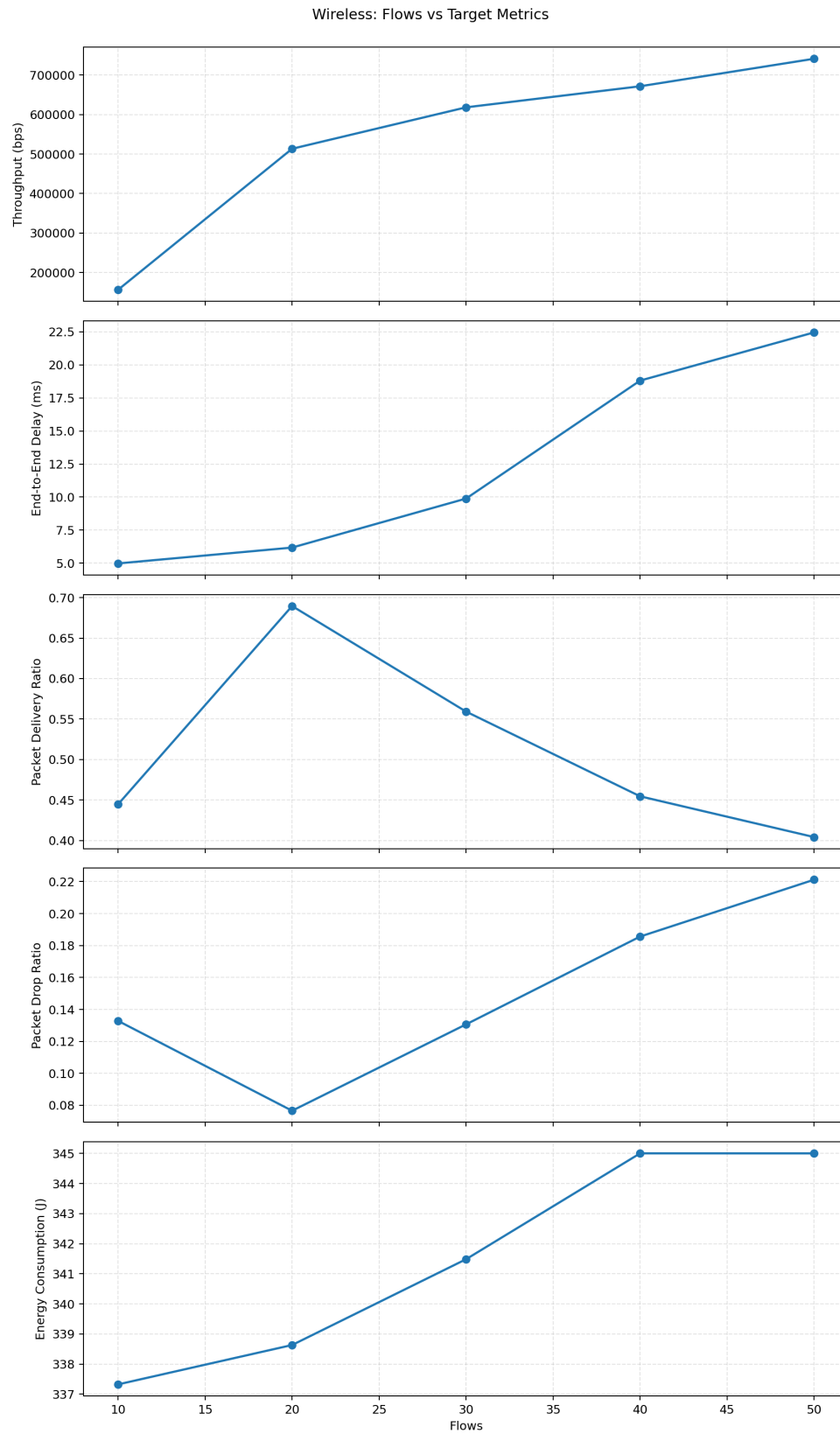


Figure 2: Wireless: Performance vs Number of Flows

Throughput peaks at moderate flow levels and decreases as contention increases. Delay rises significantly at higher flows, indicating congestion. Packet delivery ratio drops sharply, while packet drops increase.

Table 2: Wireless flow sweep results

Flows	Throughput (bps)	Delay (ms)	PDR	Drop Ratio	Energy (J)
10	156168	4.98	0.4446	0.1328	337.33
20	513432	6.18	0.6896	0.0766	338.63
30	618408	9.89	0.5592	0.1307	341.49
40	671760	18.81	0.4545	0.1856	345.00
50	741528	22.46	0.4042	0.2212	345.00

5.1.3 Effect of Node Speed

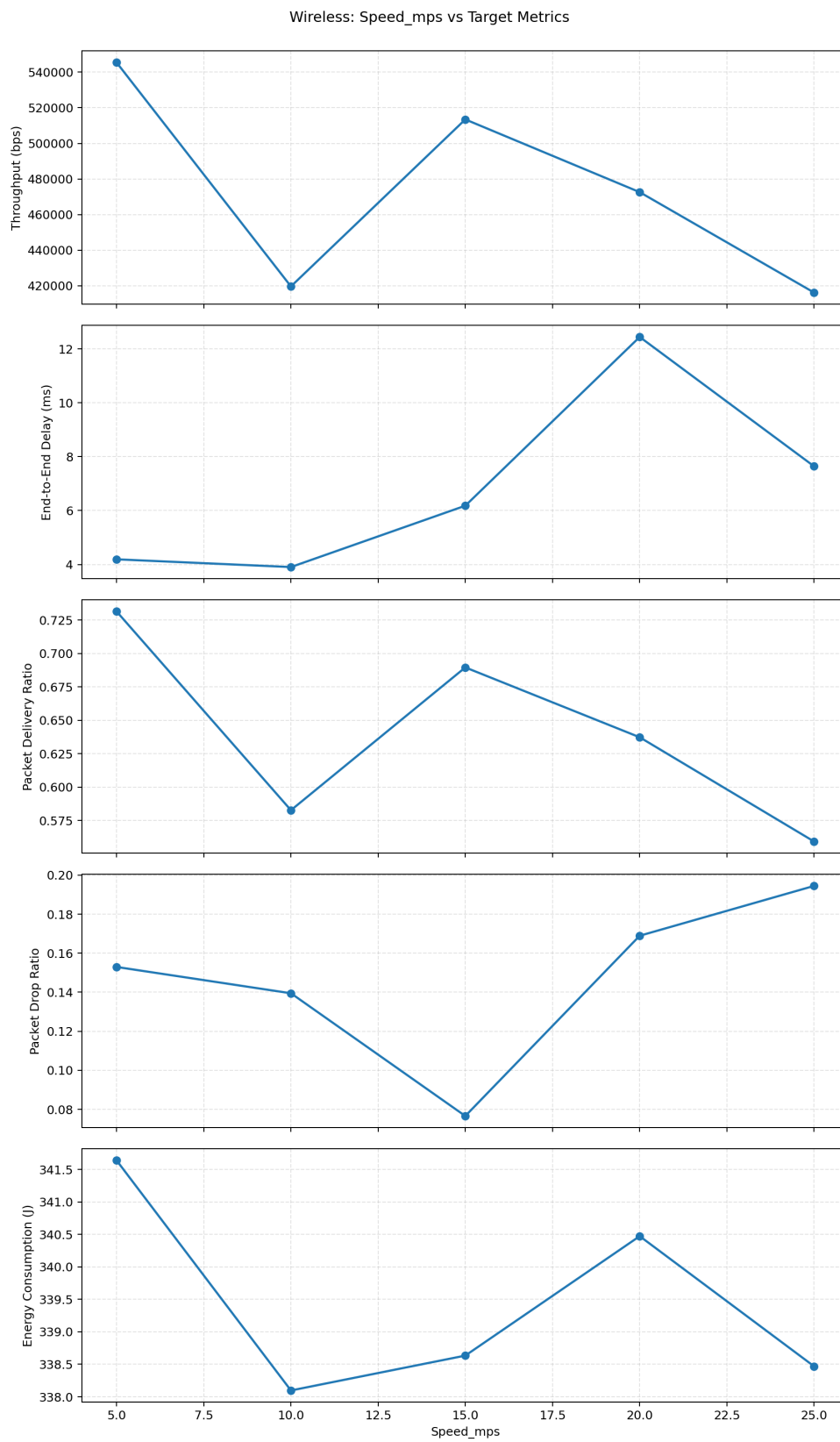


Figure 3: Wireless: Performance vs Node Speed

Higher mobility leads to unstable routes, reducing throughput and PDR. Delay and packet loss increase due to frequent route breakages.

Table 3: Wireless speed sweep results

Speed (m/s)	Throughput (bps)	Delay (ms)	PDR	Drop Ratio	Energy (J)
5	545400	4.19	0.7315	0.1530	341.64
10	419688	3.90	0.5828	0.1395	338.10
15	513432	6.18	0.6896	0.0766	338.63
20	472608	12.44	0.6373	0.1689	340.47
25	416232	7.64	0.5594	0.1945	338.47

5.1.4 Effect of Number of Nodes

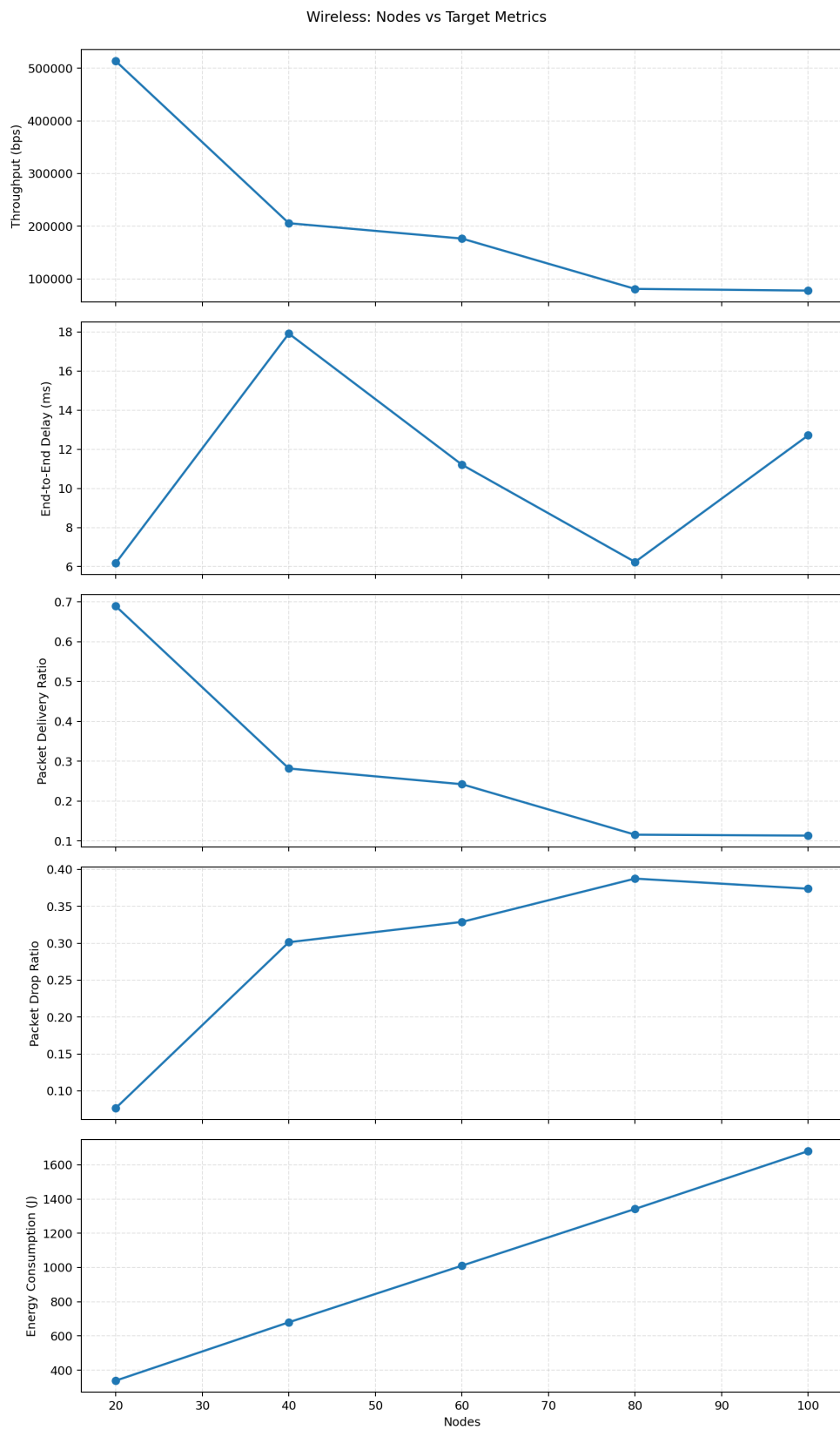


Figure 4: Wireless: Performance vs Number of Nodes

Increasing node density initially improves connectivity but eventually leads to congestion, reducing throughput and PDR while increasing delay and energy consumption.

Table 4: Wireless node sweep results

Nodes	Throughput (bps)	Delay (ms)	PDR	Drop Ratio	Energy (J)
20	513432	6.18	0.6896	0.0766	338.63
40	205848	17.92	0.2819	0.3014	679.70
60	176688	11.21	0.2424	0.3289	1010.64
80	81216	6.24	0.1157	0.3876	1342.05
100	77976	12.71	0.1133	0.3738	1680.03

5.2 Wired Network Results

5.2.1 Effect of PPS

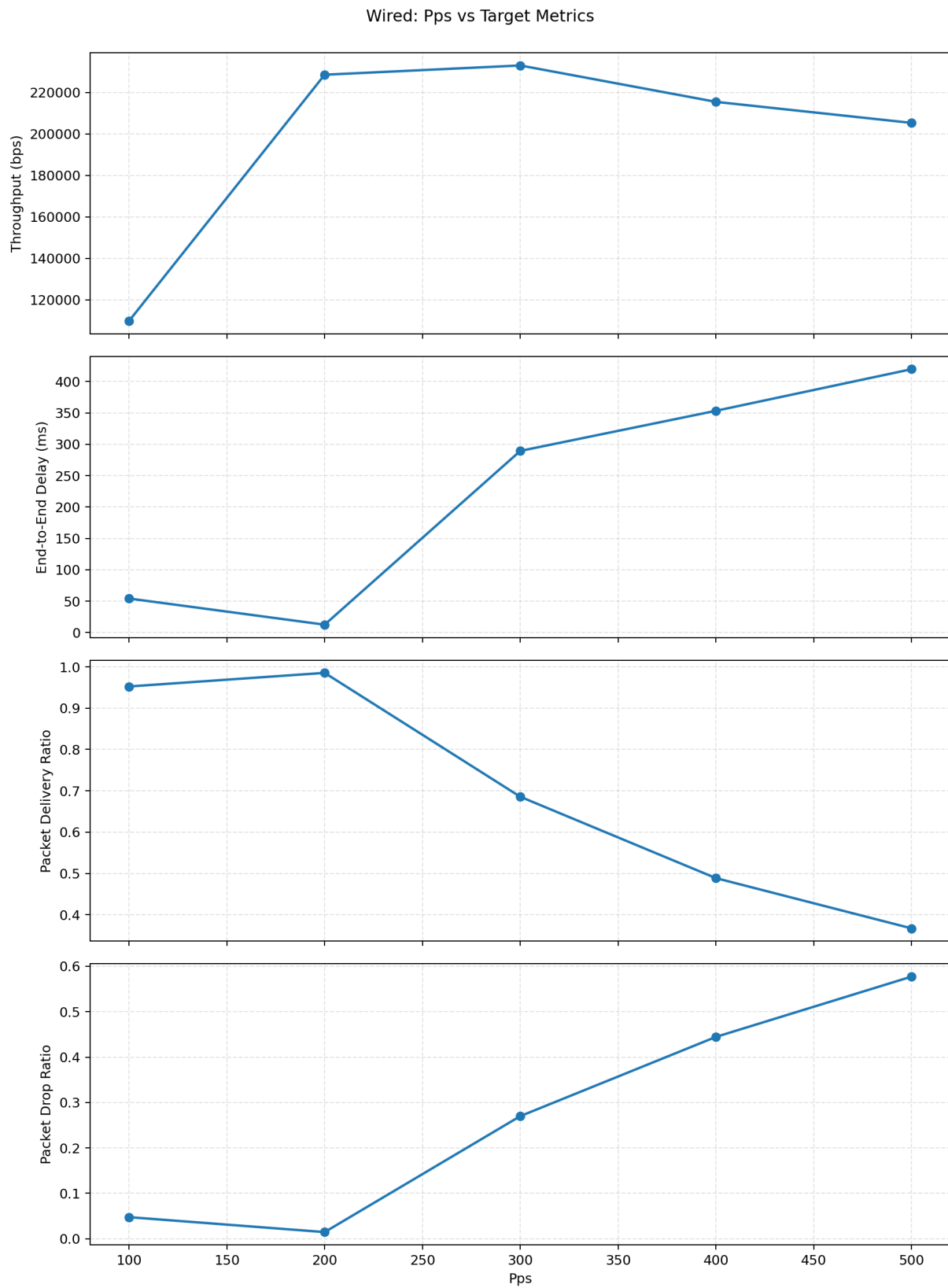


Figure 5: Wired: Performance vs PPS

Throughput increases up to a threshold and then drops sharply due to buffer overflow and congestion. Delay increases significantly at higher PPS values.

Table 5: Wired PPS sweep results

PPS	Throughput (bps)	Delay (ms)	PDR	Drop Ratio
100	109728	54.12	0.9525	0.0475
200	228600	12.39	0.9854	0.0146
300	233064	289.45	0.6855	0.2700
400	215496	353.41	0.4886	0.4445
500	205344	419.75	0.3670	0.5773

5.2.2 Effect of Number of Flows

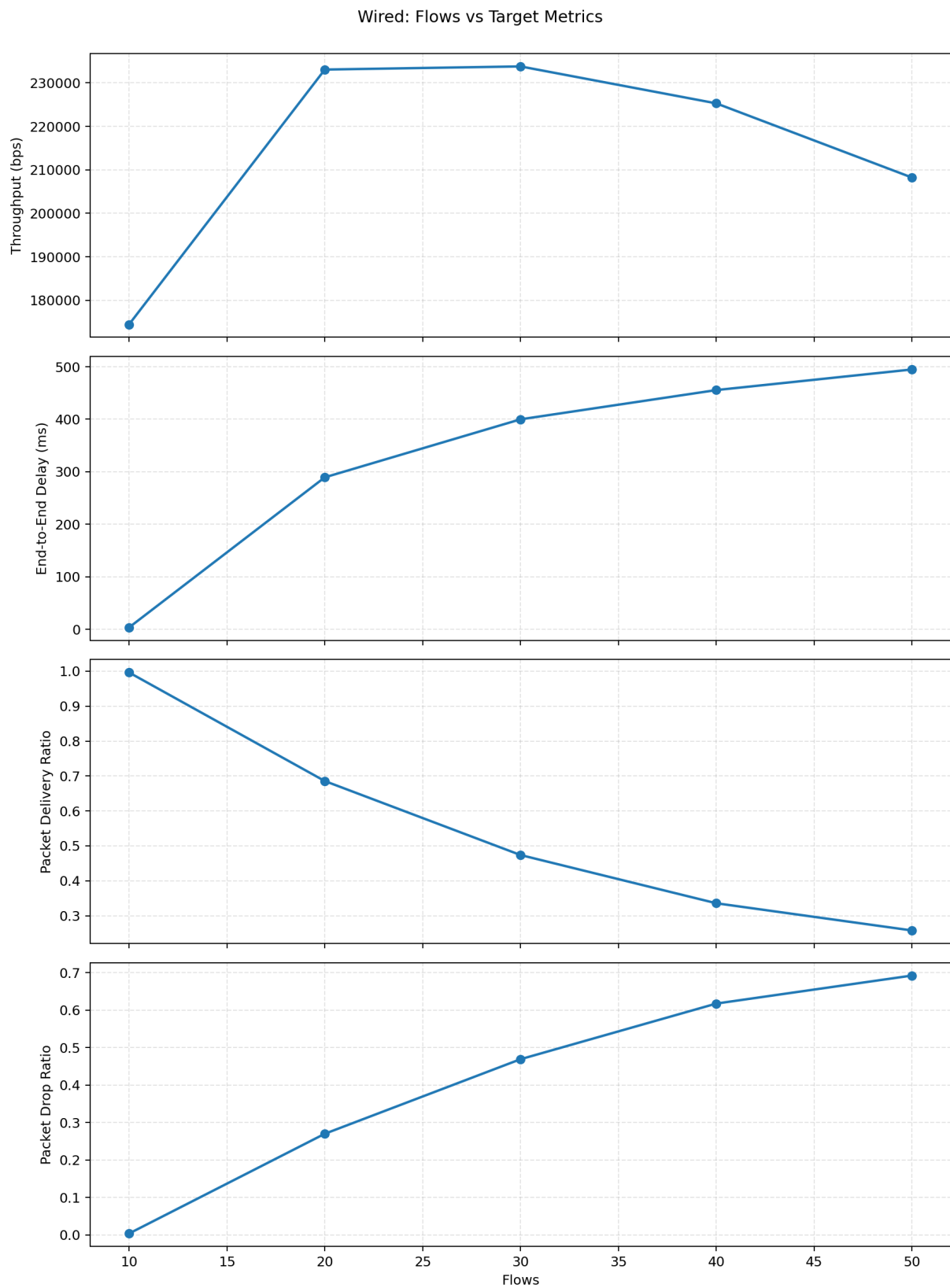


Figure 6: Wired: Performance vs Number of Flows

As flows increase, throughput decreases due to contention. Delay and packet drop ratio increase significantly.

Table 6: Wired flow sweep results

Flows	Throughput (bps)	Delay (ms)	PDR	Drop Ratio
10	174456	3.76	0.9963	0.0037
20	233064	289.45	0.6855	0.2700
30	233784	399.86	0.4737	0.4691
40	225288	455.52	0.3357	0.6176
50	208224	494.76	0.2578	0.6924

5.2.3 Effect of Number of Nodes

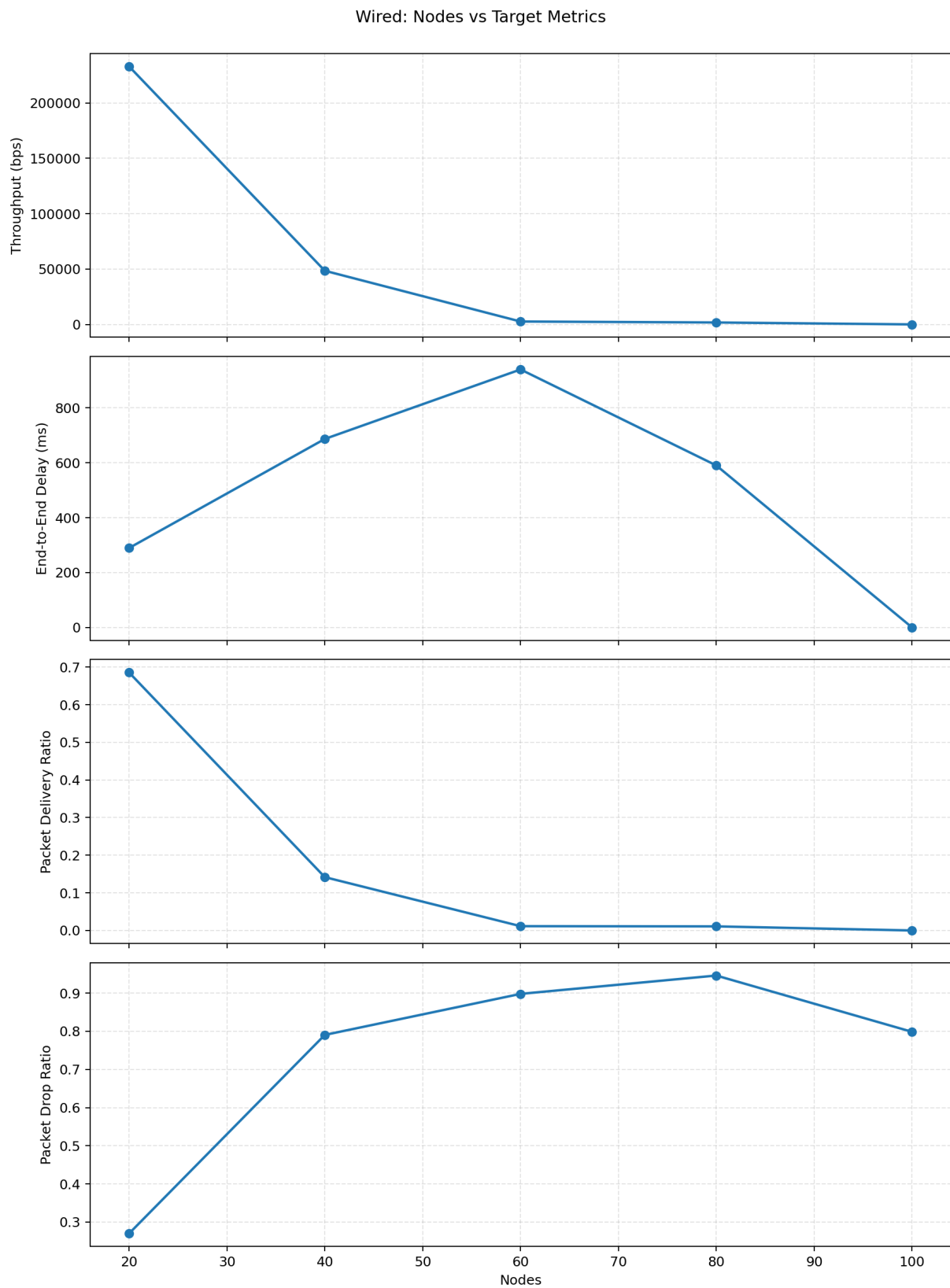


Figure 7: Wired: Performance vs Number of Nodes

Higher node counts result in severe degradation in throughput and reliability due to routing overhead and congestion.

Table 7: Wired node sweep results

Nodes	Throughput (bps)	Delay (ms)	PDR	Drop Ratio
20	233064	289.45	0.6855	0.2700
40	48456	686.84	0.1417	0.7904
60	2592	940.10	0.0114	0.8980
80	1728	590.31	0.0109	0.9460
100	0	0.00	0.0000	0.7985

6 Summary Findings

- AOMDV performs well under moderate load conditions.
- High PPS and flow counts lead to congestion and performance degradation.
- Mobility negatively impacts routing stability in wireless networks.
- Increasing nodes improves connectivity initially but degrades performance beyond a threshold.
- Wired networks show sharper performance drops under heavy load compared to wireless networks.

7 Discussion and Reflection

The results highlight the limitations of AOMDV under high traffic and dense network conditions. While multipath routing improves resilience, it introduces additional overhead that becomes significant under heavy load.

The deviation from the originally specified parameters was necessary due to:

- Hardware limitations preventing large-scale simulations
- Excessive congestion causing unstable or non-converging simulations

This adjustment ensured meaningful and stable results.

From the observations:

- There exists an optimal operating region for AOMDV
- Beyond this region, performance degrades rapidly
- Trade-offs between throughput, delay, and reliability are clearly visible

Future work could explore:

- Comparison with other routing protocols (AODV, DSR)
- Optimization techniques for congestion control
- Adaptive routing based on network conditions

8 Conclusion

This study demonstrates that AOMDV is effective in moderate network conditions but struggles under high load and dense deployments. Careful parameter tuning is essential to achieve optimal performance in both wired and wireless environments.